Master Beekeeping
Key Points

A distilled summary of the essentials from Parts I–IV

Organized by part and module for quick study and review.

Part I — Honey Bee Evolution, Biology, and Behavior

The Honey Bee Colony as a Superorganism

- A honey bee colony functions as a single 'superorganism': 30,000–50,000 individuals at summer peak that cannot survive alone but thrive as an integrated, cooperative unit.
- The queen depends on workers for nest building, defense, food, brood rearing, and fighting pests; workers depend on the queen to lay all eggs.
- Social living gives bees efficient foraging (via recruitment/communication), nest thermoregulation, and reproduction by swarming.
- Superorganisms (also termites, ants, some wasps, naked mole rats) are groups whose members function together as one larger organism.
- Social immunity—grooming, hygienic removal of diseased brood, and even 'social fever' to cook out chalkbrood—counters the high disease risk of crowded living.

Classification & Distinguishing Honey Bees

- Western honey bee = Apis mellifera ('mellifera' = honey-bearer); classified Kingdom Animalia > Phylum Arthropoda > Class Insecta > Order Hymenoptera > Family Apidae > Genus Apis > species mellifera.
- There are ~20,000 bee species worldwide and 8 honey bee species in genus Apis; only same-species bees can interbreed, but closely related species can share parasites (Varroa, Nosema originated on Apis cerana).
- Master Beekeepers must distinguish honey bees from look-alikes: yellow jackets and paper wasps (slender 'waisted', carnivorous larvae, no honey) and syrphid (hover) flies (one pair of wings, huge eyes, hover).

Evolution & Eusociality

- Bees evolved ~130 million years ago from carnivorous solitary wasps that switched to feeding young pollen—'vegetarian wasps'—coevolving with flowering plants.
- Honey bees are eusocial, the most advanced social living, defined by three traits: overlapping generations, cooperative brood care, and reproductive division of labor.
- Castes are rigid: a female develops as either queen or worker and cannot switch.
- Kin selection explains worker 'altruism': via haplodiploidy, full sisters share ~75% of genes (vs 50% in humans), so helping rear the queen's daughters spreads more of a worker's genes than reproducing herself.
- Polyandry: queens mate with ~12–50 drones, boosting colony genetic diversity, health, and productivity; worker policing (eating worker-laid eggs) suppresses worker reproduction.

- Wild bees matter: ~77% of bee species are solitary; many are important crop pollinators; several wild species are now in decline or endangered, and beekeepers are stewards/advocates for all pollinators.

Honey Bee Reproduction

- Reproduction happens at two levels: the whole colony reproduces by swarming; individuals reproduce by mating.
- Genetics basics: DNA coils into 16 chromosomes; the honey bee genome (~10,000 genes, sequenced 2006) is ~1/10 the size of the human genome; beekeepers can selectively breed multi-gene traits (temperament, hygiene, honey production).
- Mating occurs at drone congregation areas (DCAs) 50–250 ft up; queens take 1–3 flights, mating with ~10–20 drones; the drone's endophallus ruptures and he dies (leaving the 'mating sign').
- A mated queen stores ~5–6 million sperm for life and lays ~1,000–1,500 eggs/day; she controls fertilization—fertilized eggs become females, unfertilized become drones.
- Queen fertility is signaled by 'queen pheromone'; when it weakens or disappears, workers build supersedure or emergency queen cells. Queenless workers can become laying workers (drone eggs, multiple per cell, on cell sides).
- The stinger is a modified ovipositor—only females can sting.

Honey Bee Development

- All castes spend 3 days as egg + 6 days as larva; pupal time differs—queen ~7–8 days, worker ~12, drone ~15 (queen ~16 days total to emergence; worker ~21; drone ~24).
- Fast queen development is favored: first queen out kills rivals and gets a colony head start.
- Sex is set by the Complementary Sex Determination (CSD) gene: two different alleles = female; one allele (or two identical) = male. Identical alleles make 'diploid males' that workers cannibalize—inbreeding causes spotty brood.
- Caste is environmental, not genetic: any young female larva (<36 hrs old) can become queen or worker. Royal jelly quantity/duration drives it via juvenile hormone and vitellogenin.
- Grafting older larvae can yield poor 'intercastes' (mix of queen/worker traits).
- Workers shift jobs with age (polyethism): cell cleaning → nursing/queen tending → comb building/food handling → guarding/fanning → foraging; the timing is flexible and responds to colony needs.
- Lifespan: summer workers 2–5 weeks; spring/fall workers 1–2 months; winter bees 4–5 months; queens 2–3 years; drones evicted before winter.

How Honey Bees Communicate

- Bees communicate by chemicals (pheromones), touch/vibration, and a little vision—mostly in the dark.

- Queen pheromone (esp. queen mandibular pheromone, QMP) is spread by the retinue via trophallaxis/antennation; it signals the queen's presence, inhibits worker ovaries and queen rearing, and stabilizes the colony.
- Brood, worker (Nasonov—orientation/aggregation; lemongrass oil mimics it), and alarm pheromones (smells like banana—isoamyl acetate) coordinate colony behavior.
- The waggle dance (von Frisch; studied by Cornell's Tom Seeley) encodes direction (angle relative to vertical/sun) and distance (waggle-run duration) to food; more circuits = better source.
- Other signals: shaking signal ('shake and wake'/switch tasks), tremble dance (recruit nectar receivers), and the 'stop' signal (beeping to halt dances for a failing source).

Honey Bee Biology (Body Systems)

- Nervous system: brain + ganglia; ~1 million neurons let bees learn/remember smells, count to four, grasp 'zero'; the mushroom body (20% of brain) is the learning/memory center.
- Senses: mechanoreception (touch, gravity via hairplates, hearing via subgenual/Johnston's organs); smell is the most important sense (170 olfactory receptors—more than fruit fly or mosquito); weak taste; UV/blue/green color vision (can't see red), detects polarized light to navigate by the sun.
- Vision tips for beekeepers: use distinctive landmarks and spaced/varied hives to reduce drift and disease spread; move hives <2 miles only in small steps or over winter.
- Flight: ~240 wingbeats/sec, up to ~20 mph; four wings hooked by hamuli.
- Open circulatory system: 'hemolymph' (blood+lymph) is pumped by a heart in the abdomen; carries nutrients, waste, hormones, immunity.
- Respiration via tracheae/spiracles (tracheal mites clog these). Digestion: crop ('honey stomach') stores nectar; midgut (ventriculus, with peritrophic membrane) is where Nosema/viruses attack; bees hold waste to take cleansing flights (dysentery inside hive signals problems).

How Honey Bees Construct & Maintain Nests

- Comb is built from wax (energetically costly: ~5–20 lb honey per 1 lb wax), produced by workers ~12–18 days old; building needs both excess nectar and a need for space.
- Propolis (plant resin + wax) seals and insulates the nest and provides antimicrobial protection; recommend rotating out the 2 oldest/darkest combs yearly (wax accumulates pesticides and pathogens).
- Bee space = 3/8 inch between combs; this principle underlies Langstroth's 1852 movable-frame hive that revolutionized beekeeping.
- Nest layout: brood in a sphere low/center, surrounded by pollen then honey arcs; watch for honey-bound/pollen-bound colonies (can trigger swarming); reverse hive bodies in spring to give space.
- Preferred wild nest: ~10-gallon cavity, small south-facing entrance high off the ground—aids defense and thermoregulation; bait hives with old comb attract swarms.

- Thermoregulation: brood kept ~92–97°F. Heating—winter cluster shivering (clusters below ~57°F); don't open brood nest in cold (<59°F). Cooling—dispersing, fanning, evaporative water cooling, and bearding.

Part II — The Science and Art of Beekeeping

Safety & Biosecurity in the Bee Yard

- Some stings are inevitable; protective gear is a personal trade-off between protection and mobility. Warn and equip visitors. Colonies that sting excessively may need moving, requeening, or culling.
- Sting reactions: local (pain/swelling/itch 2–3 days; treat with ice and oral antihistamines) vs systemic/allergic (within minutes–1 hr, beyond the sting site—can be life-threatening). Don't restrict blood flow; age affects allergy risk.
- Practice hygienic management: space hives and use landmarks to reduce drift, wash hands and sterilize the hive tool between yards—drifting and robbing spread Varroa, viruses, Nosema, and American foulbrood.
- Robbing (bees stealing honey from other colonies) peaks in dearth/late summer and can kill a weak colony in a day. Stop it fast by reducing the entrance to 1–2 bee-widths; prevent it with strong queenright colonies and careful feeding.
- Honey bees are managed livestock; minimize competition and disease spillover to ~4,000 native US bee species (~450 in NY).

Inspecting the Colony

- Inspect every 10–14 days in spring/summer (less in fall/winter); keep each inspection to ~5–10 minutes to limit disturbance.
- Check population, queen status (worker eggs? supersedure/swarm cells? laying workers?), and brood quantity/quality (solid vs spotty, all ages present).
- Read the outside first: predator signs (skunk scratching, bear damage), entrance activity, and temperament before opening. Work top box down to reduce aggression; pull ~2–3 frames per super and ≥4 per brood box.
- A colony assessment is a deeper count (frames of bees, brood by stage, honey, pollen, drawn comb) used to quantify growth/productivity.
- Keep good records (ID every colony/yard); emerging tools include precision/'smart' beekeeping sensors and AI—useful but interpret indirect signals (e.g., Varroa sensors) with caution.
- Lifting tip: lift frames with finger strength, not your back, to avoid injury.

Honey Bee Nutrition

- Bees need carbohydrates (nectar/honey) year-round and protein/fat/vitamins/minerals (pollen/bee bread) mainly for brood and young bees; one colony may visit 200+ plant species a year.
- Nectar sugar concentration (5–92%; sucrose/glucose/fructose) guides foraging—bees prefer higher sugar. Learn your region's key honey and pollen plants (early bloomers like dandelion are valuable despite lower quality).

- Feed carbs when stores are low (winter/early spring), after installing packages/nucs/swarms, during dearth, when requeening, or to draw comb. Use 1:1 syrup to stimulate brood/comb; 2:1 for winter stores.
- In winter feed solid (dry) sugar, not syrup—excess water can cause dysentery, and dry sugar won't prematurely stimulate brood rearing.
- Only feed honey from your own disease-free colonies (store-bought/other beekeepers' honey can carry AFB or Nosema spores). Fresh pollen > pollen substitute; feed protein in late winter/early spring to build population before nectar flows.

Managing Colony Population (Splits, Nucs, Merges)

- Colony population in the Northeast cycles from ~5,000–10,000 in winter to a 40,000–50,000 peak in mid-June, then declines; understanding this cycle underpins all management.
- Split strong colonies (≥6 frames brood, ≥4 frames food) to replace winter losses (~17% considered acceptable) and grow your operation affordably. Move ~2–3 brood frames + 2 food frames into the new box.
- Nucleus colonies (nucs, 2–5 frames) are valuable for spare resources, mating reared queens, and queen introductions; keep 'resource nucs' on hand. Overwintering summer-made nucs also breaks the brood cycle and lowers Varroa.
- Merge weak colonies (<8 frames of bees) into stronger ones—but first rule out disease. Use the newspaper method so odors mingle and bees accept each other; keep only one queen.

Swarming

- Swarming is natural colony-level reproduction: 1/2–2/3 of the bees plus the old queen leave; the parent colony requeens itself (≈6 weeks to rebuild brood).
- ~80% of Northeast swarms issue mid-May to mid-July, triggered by congestion (crowded adults/brood); warm calm days after bad weather are 'swarmy'. A second smaller season runs mid-Aug to mid-Sep.
- Swarming has upsides (requeening, brood break that lowers mites, leaving behind toxins/pathogens) but costs honey, risks the parent colony, and spreads disease to neighbors and wild bees.
- Prevention targets the future signals (mainly congestion—add space, reverse boxes, split). Swarm control acts once swarm cells exist (remove old queen, cut cells, etc.). Ineffective: clipping the queen's wing or putting an excluder below the brood nest.
- Don't tell swarming from orientation flights (bees returning) or robbing (bees fighting to get IN). The Dyce Lab recommends preventing swarms rather than letting colonies swarm loose. Bait hives mimic preferred nest sites to catch swarms for free.

Queen Management

- The queen drives colony performance; queen failure rates have risen (surveys report up to ~50% replaced), so evaluating and replacing queens is a core skill.

- Queenless signs: 'roaring' (loud agitated fanning) within an hour as QMP disappears, then emergency cells, eventually laying workers. Time matters when requeening—too little time and they can't make a queen; too much and laying workers block acceptance.
- ~30% of introduced queens die within 6 months. Boost acceptance by adding a frame of young brood/eggs with a new caged queen (research shows ~200% improvement). Packages are hardest (unrelated bees).
- Routine requeening (spring or late-summer/fall) improves health/productivity; fall requeening also interrupts the brood cycle. Two-queen colonies can boost honey but are labor-intensive (better for hobbyists/sideliners).
- Favor locally adapted / 'survivor' stock (thrived ≥2 years/winters with minimal intervention)—often better suited to your climate.

Variation Within Apis mellifera

- After evolving in Asia, honey bees formed several genetically distinct Old World lineages (e.g., A=African, M=W/N European, C=E European, O=W Asian).
- North America: German black bees (M) arrived 1622; Italians (1859), Carniolans, Caucasians and others followed—shaping today's stock.
- Africanized honey bees ('killer bees') are A. m. scutellata hybrids accidentally released in Brazil (1956); highly defensive, they spread north reaching Texas (1990) and now occupy parts of the southern US.

Queen Rearing (Doolittle Method)

- Rearing your own queens makes you self-sufficient. Simplest: let a strong nuc (with eggs) or one with swarm cells finish raising queens.
- Timing is critical—start from eggs/larvae <36 hours old; swarm season is easiest (abundant food, rising population, available drones).
- Grafting (transferring young larvae into artificial cups) gives full control; expect ~80–85% success after practice (~50 tries). Avoid injuring/drying larvae.
- Use cell builders: a cell starter (queenless, royal-jelly-rich, larvae 1 day) then a cell finisher (queenright, very populous) to draw and cap cells. Move near-emergence cells to mating nucs.
- Schedule (Day 1 = egg laid): queen emerges ~Day 16, matures ~1 week, mates, and lays ~2 days after her mating flight.

Breeding for Desired Traits

- Breeding increases desirable heritable traits (gentleness, good laying/brood pattern, honey production, hygienic behavior, mite resistance) and removes undesirable ones—select queen-mother and drone-mother colonies from your best.
- Genetic diversity (queens mating with many drones) improves disease resistance and adaptability—an alternative to narrow single-trait selection.

- Avoid inbreeding: it loses diversity, raises diploid-male/spotty-brood risk, and leaves colonies unable to cope with new stresses.
- Drone selection matters as much as queen selection (females get half their genes from the father). Add drone comb to drone-mother colonies 35–40 days ahead.
- Controlled mating (isolated yards ≥9 miles out, or instrumental insemination) guarantees parentage; open mating is easier but less certain. Evaluate colonies in a testing apiary, scoring multiple traits and minimizing drift.

Part III — Managing Pests and Diseases

Honey Bee Immunity

- Individual defenses: the sting, grooming, and protective body layers (cuticle, peritrophic membrane) keep invaders out.
- Bees have only an innate immune system (no antibodies/memory cells, fewer immune genes than other insects): hemolymph cells engulf pathogens (phagocytosis/nodulation) and bees make antimicrobial compounds fast.
- Social immunity is the most powerful layer: grooming, hygienic removal of diseased brood, social fever, and propolis (antimicrobial; bees even 'entomb' dead intruders).
- A healthy gut microbiome (acquired from nestmates/hive) outcompetes harmful bacteria.
- When natural defenses aren't enough, the beekeeper intervenes using Integrated Pest Management (IPM)—the framework for the whole course.

Insect Pests (Beetles, Moths, Wasps)

- Wax moths are a secondary pest—mostly damaging weak/queenless colonies or stored equipment (gray webbing, tunneled comb, frass). Prevent by keeping colonies strong and right-sized; treat stored equipment by freezing/heating (no chemical treatment for occupied hives).
- Small hive beetle (SHB, Aethina tumida; in US since 1998) mainly harms weak colonies, fermenting honey into a slimy mess. Adults fly up to 8 miles and survive ~10 days without food. Manage with beetle traps (oil/vinegar) first, then pesticides/ground drenches; keep apiaries/honey houses clean.
- Yellowjackets attack in late summer/fall, overwhelming weak colonies—reduce robbing and set wasp traps.
- Two invasive hornets that decapitate adult bees and raid brood have been detected in North America; they recruit nestmates by pheromone and can devastate colonies.

Fungal Enemies — Nosema

- Nosema is a gut fungal/microsporidian parasite of adult bees—common worldwide but not a major US loss driver; almost all US infections are now Nosema ceranae (N. apis is rare).
- Symptoms aren't unique (dwindling, reduced honey, supersedure, winter loss); only microscopy or molecular tests diagnose it. Send ≥100 live bees to the USDA Beltsville Bee Lab (free).
- Most common in spring (~86% of NY colonies; ~0.87M spores/bee), much less in autumn; pesticide exposure (e.g., imidacloprid) worsens vulnerability.
- Prevent with good drainage/sun/ventilation and spring nutrition; spores persist in comb for years. Many colonies self-resolve by autumn. Fumagillin (Fumagilin-B) is NOT reliably effective and may worsen N. ceranae; treatment threshold often cited around 1M spores/bee.

Varroa Mites (the most destructive parasite)

- Varroa destructor are sesame-seed-sized reddish-brown ectoparasites that jumped from Apis cerana to Apis mellifera (1960s); reached US Florida in 1987 and spread nationwide by the mid-1990s (migratory beekeeping accelerated it).
- Life cycle: reproductive stage in capped brood (prefers drone cells) + phoretic stage riding adults (often nurse bees). Mites feed mainly on the fat body, depleting lipids needed for winter bees.
- Mites vector 8+ viruses (esp. deformed wing virus); the mite+virus combination is far deadlier than either alone. Advanced infestation = Parasitic Mite Syndrome (high mites, deformed-wing adults, spotty brood). Mite populations peak in late summer/autumn (monitor Aug–Oct).
- Untreated colonies usually die within 6 months–2 years; Varroa accounts for >85% of overwintering losses in one large study. You can't eradicate it—keep levels low year-round via IPM.
- Monitor monthly (Mar–Oct) with the only reliable methods: powdered sugar shake or alcohol wash (sticky boards, drone uncapping, ether roll are unreliable for counts). Treatment thresholds ≈ 2 mites/100 bees (Apr–Jul) and 3/100 (Aug–Oct).
- Genetic resistance is the best head start: VSH, Minnesota Hygienic, Russian, Purdue Ankle Biters, Saskatraz; hygienic behavior (freeze-kill test: >90% removal = hygienic) and grooming reduce mites.
- Cultural (non-chemical) methods—drone-brood trapping (freeze frames), screened bottom boards, small hives, colony spacing, brood interruption (swarm/split)—each help but only work combined in IPM. Small-cell foundation does NOT work.
- Chemicals (8 registered in NY) fall into synthetics, essential oils, and organic acids. Amitraz (Apivar) was the backbone but widespread resistance (2024/25) is driving record losses; formic acid (Mite Away Quick Strips) is honey-super-safe; Apistan/CheckMite+ lost efficacy to resistance. Always follow the label and use PPE—even 'natural' treatments can harm bees/people. Plan a Varroa management calendar in advance.

Viruses

- Apis mellifera hosts 19+ viruses; effects range from asymptomatic to colony death, and they're the hardest diseases to diagnose.
- Deformed wing virus (DWV) is the most prevalent and one of the most destructive (Varroa-transmitted; found in 80–96% of colonies)—deformed wings, dead pupae, bloated dark abdomens.
- Other key viruses: sacbrood (oldest known, 1913; curled larva, fluid sac), black queen cell virus (2nd most prevalent, usually asymptomatic), the acute paralysis complex (ABPV/KBV/IAPV—Varroa-transmitted paralysis), and chronic bee paralysis (hairless-black/paralysis forms).
- No treatments target viruses—management = keep colonies healthy (nutrition, strong population, fertile queen) and, above all, keep Varroa low to limit virus transmission.

Other Parasites

- Tracheal mites (Acarapis woodi) infest the breathing tubes ('Acarine disease'); caused massive early-20th-century losses but are minor today. Signs: crawling bees that can't fly, K-wing, failure to cluster. Diagnose by microscope; manage with resistant stock (Buckfast, NW Carniolan, Russian) and grease patties.

- 'Zombees' (honey bees parasitized by phorid flies that drive them to lights at night) are uncommon in the US and are NOT the cause of Colony Collapse Disorder.

Brood Diseases

- Chalkbrood (fungus Ascosphaera apis): larvae turn into hard white/then brown-black chalky 'mummies'; usually minor, managed by strong/hygienic colonies and good ventilation.
- American foulbrood (AFB, Paenibacillus larvae) is the most serious bacterial brood disease—spores survive 40+ years. Signs: spotty brood, sunken/perforated cappings, coffee-colored larvae that 'rope' out ~1 inch, foul odor. Inspect 3×/year. Confirm with Vita test kit or Beltsville lab.
- AFB is reportable by law in many areas (e.g., NY) and infected colonies are typically destroyed by burning under inspector oversight; disinfect equipment and don't share/buy used gear. Antibiotics only kill vegetative bacteria (not spores) and now require a veterinary prescription/VFD; prophylactic use is no longer recommended due to resistance.
- European foulbrood (EFB, Melissococcus plutonius): twisted brown/yellow larvae with visible trachea, sour odor; less severe than AFB—does NOT require burning; can be requeened and/or treated. Test to distinguish from AFB.
- Idiopathic Brood Disease Syndrome ('snotty brood') can be confused with foulbroods/PMS—test to confirm.

Bee Health — The Big Picture

- US/Canada/Europe report ~30–45% annual colony losses since ~2010; 53 Northeastern wild bee species are also declining.
- Colony Collapse Disorder (CCD, from 2006): sudden disappearance of adults leaving brood/food behind—a specific syndrome, NOT the same as general annual losses; no single cause was found (multiple interacting factors).
- Current losses aren't historically unique—bee numbers have risen and fallen with economic, social, and environmental factors.
- Be the colony's 'primary care physician': inspect, monitor, use the symptom chart, and communicate with other beekeepers to diagnose problems.
- Pathogen spillover: keeping honey bees healthy protects wild bees, since shared pathogens (Nosema ceranae, DWV, BQCV, etc.) can spread at shared flowers.

Part IV — The Rewards and Contributions of Beekeeping

Honey

- Honey is the colony's main energy source—mostly fructose and glucose; bees ripen nectar by adding enzymes and reducing moisture below ~18%.
- Composition: ~98% of solids are sugars (avg ~17% moisture, ~38% fructose, ~30% glucose, plus acids, vitamins, minerals). Honey can carry Clostridium botulinum spores—never feed honey to infants under 1 year.
- Medicinal value: honey helps with coughs and wound care; manuka honey (from NZ/Australia) is exceptionally antibacterial due to methylglyoxal (MGO)—used in some hospital wound dressings.
- Over time honey crystallizes (glucose precipitating—rate depends on nectar source, particles, temperature) and can ferment if moisture >18–19% (yeasts turn sugar to alcohol/CO_2).
- Processing: raw honey is just extracted/strained (no heat); straining/filtering removes particles/pollen; pasteurization delays crystallization but destroys natural enzymes. Beekeepers with >50 colonies typically need a bee-proof honey house (concrete floors/drains; tangential vs radial vs larger extractors).
- Selling: small-scale via local retail; sideliners/commercial often sell wholesale to packers/brokers. Honey price fluctuates with supply/demand of domestic vs imported honey; all labels must meet food-labeling laws, and USDA has voluntary color/clarity grade standards.

Additional Hive Products

- Beyond honey: beeswax, pollen, propolis, royal jelly, venom, and mead—using hive products in human health is called apitherapy.
- Beeswax (rendered from cappings; melts ~143–149°F, discolors above 185°F; 'slumgum' = filtered debris) makes superior candles (burn longer, less smoke than paraffin) plus cosmetics, polish, and food wraps.
- Pollen is a protein-rich supplement (sold as pellets); harvest temporarily with pollen traps so you don't starve the colony; no clinical evidence it treats conditions (including allergies).
- Propolis (antibacterial/antifungal resin) is harvested by scraping or traps and sold as tinctures/capsules; royal jelly commands the highest price (nutrition/cosmetics); bee venom (collected by electroshock) is used to treat sting allergies and has anti-inflammatory effects.
- Mead is fermented honey (honey + water + yeast)—dry to sweet, still or sparkling; flavor varies with the honey used.

Pollination Services

- Bees are the world's main insect pollinators—wild and managed bees contribute ~$20 billion to US agriculture and $217+ billion worldwide.

- Pollination is plant sexual reproduction; many staples (wheat, corn, rice) are wind-pollinated, but many fruits/vegetables/nuts depend on bees.
- A pollination-ready colony should be strong and growing—at least ~8 frames of bees and 4 frames of brood; preparing strong spring colonies starts the previous autumn.
- Hive stocking rates are crop-specific (some crops need more hives due to low-attraction flowers like alfalfa). Pollination rentals have become a major income source—often exceeding honey for commercial beekeepers—but are stressful on bees.
- Price rentals to be profitable (mileage, labor, trucking, feed, risk). Use a written pollination contract. Transport requires permits/certifications, screened entrances, ventilation/water, secured loads, and frequent monitoring; manage 'hitchhiker' bees at stops.

Communicating About Pesticides

- Pesticides (insecticides, herbicides, fungicides) contain active + 'inert' ingredients—both can affect bees; ~1 billion lb are applied yearly on US farmland (mostly herbicides).
- Harm is measured via acute toxicity (LD50 = dose killing half a population), residual and chronic toxicity, and sublethal/synergistic effects.
- Insecticides are generally the most toxic to bees; major synthetic classes: organophosphates, carbamates, pyrethroids, insect growth regulators, and neonicotinoids.
- Neonicotinoids (from the 1990s) are now the world's most-used insecticide; nitro-types (imidacloprid, clothianidin, thiamethoxam) are more toxic to bees than cyano-types—they pose real risk in some regions/conditions (Cornell's 2020 review synthesized 400+ studies).
- Herbicides (esp. glyphosate/Roundup) generally pose low direct risk but harm bees indirectly by removing floral resources. EBI fungicides (e.g., prochloraz) are dangerous synergistically—amplifying insecticide/miticide toxicity; fungicides are among the most common hive residues.
- Reducing exposure takes collaboration among beekeepers, growers, applicators, the public, and policymakers; beekeepers can cut their own miticide residues via IPM.

Turning Bees Into Business

- Grow with a clear goal/plan; expanding slowly (e.g., from a couple of colonies to 20+ over 3–6 years) is less risky than buying many colonies at once.
- Keep financial records alongside colony records; business growth has three phases—startup, market establishment, and stabilization.
- The hobby-to-business shift requires changes in time and expense management ('time is money'); research large equipment purchases carefully.
- Choose a business structure based on local laws and where products are distributed; many support programs exist (e.g., NYS Beekeeper Tech Team for colony health; USDA Farm Service Agency loans/grants for beginning and socially disadvantaged farmers).

Being a Leader in Your Community

- Mentoring new beekeepers reinforces your own skills and requires beekeeping, teaching, and communication ability—plus a realistic time commitment.
- Master Beekeepers serve through community service: lectures, workshops, school visits, media engagement, and working with researchers and government.
- Participate in research/citizen science—universities regularly seek beekeepers for pollinator-health projects.
- Build presentation skills (simple slides, compelling info, clear delivery); the Cornell program's final requires a 15-minute presentation.

Keeping Up With Science

- Three information avenues: popular science articles, primary literature, and review papers/experts; popular media often sensationalizes—trace claims back to the original study.
- Access reputable papers free via open-access journals (e.g., PLoS ONE), PubMed ('open access[filter]'), and Google Scholar; trust peer-reviewed sources.
- Read papers by their standard structure (Abstract, Intro, Methods, Results, Discussion) and interpret critically rather than getting lost in detail.
- Distinguish science from pseudoscience: science follows observe→hypothesize→test→refine; pseudoscience rests on fallacious arguments (rare in reputable peer-reviewed journals).
- Beekeepers are encouraged to run their own controlled experiments (treatment vs control groups) in their bee yards to test questions.